Solubility

CHEMISTRY REFERENCE

cations ↓ \ anions →	NO_3^- nitrate	$C_2H_3O_2^-$ acetate	ClO_3^- chlorate	Cl^- chloride	Br^- bromide	I^- iodide	SO_4^{2-} sulfate	OH^- hydroxide	CO_3^{2-} carbonate	PO_4^{3-} phosphate	S^{2-} sulfide
NH_4^+ ammonium	S	S	S	S	S	S	S	S	S	S	S
Na^+ sodium	S	S	S	S	S	S	S	S	S	S	S
K^+ potassium	S	S	S	S	S	S	S	S	S	S	S
Mg^{2+} magnesium	S	S	S	S	S	S	S	I	I	I	X
Ca^{2+} calcium	S	S	S	S	S	S	*SS*	*SS*	I	I	X
Sr^{2+} strontium	S	S	S	S	S	S	I	*SS*	I	I	X
Ba^{2+} barium	S	S	S	S	S	S	I	S	I	I	X
Al^{3+} aluminum	S	S	S	S	S	S	S	I	X	I	X
Mn^{2+} manganese(II)	S	S	S	S	S	S	S	I	I	I	I
Fe^{2+} iron(II)	S	S	S	S	S	S	S	I	I	I	I
Fe^{3+} iron(III)	S	S	S	S	S	X	S	I	X	I	X
Co^{2+} cobalt(II)	S	S	S	S	S	S	S	I	I	I	I
Ni^{2+} nickel(II)	S	S	S	S	S	S	S	I	I	I	I
Cu^{2+} copper(II)	S	S	S	S	S	X	S	I	I	I	I
Zn^{2+} zinc	S	S	S	S	S	S	S	I	I	I	I
Ag^+ silver	S	*SS*	S	I	I	I	*SS*	I	I	I	I
Hg_2^{2+} mercury(I)	S	*SS*	S	I	I	I	I	I	I	I	I
Pb^{2+} lead(II)	S	S	S	*SS*	*SS*	I	I	I	I	I	I

S soluble (more than 1 g per 100 mL)
SS slightly soluble (0.1–1 g)
I insoluble (less than 0.1 g) — a precipitate
X reacts with water or does not form
Water at about 25 °C.

Rules

1. Compounds of Group 1 ions and NH_4^+ are soluble. So are all nitrates and chlorates, and nearly all acetates.
2. Chlorides, bromides and iodides are soluble, except with Ag^+, Hg_2^{2+} and Pb^{2+}.
3. Sulfates are soluble, except with Sr^{2+}, Ba^{2+}, Pb^{2+} and Hg_2^{2+}. Ca^{2+} and Ag^+ sulfates are slightly soluble.
4. Hydroxides are insoluble, except with Group 1, NH_4^+ and Ba^{2+}. Ca^{2+} and Sr^{2+} hydroxides are slightly soluble.
5. Carbonates, phosphates and sulfides are insoluble, except with Group 1 and NH_4^+.